ACTION PLAN

S4G vNext Clean-Galaxy and Controlled-Injection Library

A reproducible, Haigh-style validation framework for foreground-object masking and bar-profile recovery

Purpose: Define the next implementable version of the S4G foreground-masking test framework
Primary users: Researcher/developer; later collaborators and reviewers
Recommended pilot: 40 clean S4G bases × 25 controlled realizations = 1,000 paired cases
Planning horizon: 12-week pilot, followed by a scale-up decision
Status: Proposed action plan

> **Recommendation** Build S4G vNext around immutable, visually validated S4G galaxy bases and independently generated foreground layers. Preserve exact truth at every stage, and evaluate parameter sets by galaxy-held-out cross-validation. Do not use APOD or processed astrophotography as quantitative training data.

1. Decision and intended outcome

S4G vNext should convert the current collection of clean galaxies and toy injections into a versioned scientific dataset and repeatable evaluation pipeline. The goal is not simply to accumulate more images. The goal is to create experiments in which the uncontaminated galaxy, added foreground signal, expected foreground mask, and resulting profile damage are all known and auditable.

The design follows the strongest principle in Haigh et al.: quantitative optimisation requires simulated or controlled data with known ground truth, while real survey fields remain valuable for qualitative stress testing. For S4G, the controlled unit should be a real 3.6 µm galaxy image plus a synthetic, separately stored contaminant layer.

Success at the end of the pilot

- A locked v1.0 catalogue of 40 clean, eligible S4G base galaxies with documented review evidence.
- A reproducible generator producing 1,000 paired clean/contaminated cases and exact pixel-level truth.
- Galaxy-held-out train, validation, and test folds that prevent multiple realizations of one galaxy leaking across folds.
- A common evaluator for SEP and MTObjects, including foreground recovery, clean-galaxy damage, and bar-profile fidelity.
- A release manifest, configurations, checksums, QA reports, and methods note sufficient to rerun the experiment.

2. Scope and scientific design

In scope

- Existing S4G 3.6 µm FITS images and the currently validated 40-galaxy clean list.
- Empirical or physically parameterised IRAC-like foreground point sources, including ordinary stars, bright cores, extended wings, and diffraction-spike cases.
- Cases sampled both randomly and deliberately across scientifically sensitive zones: bar major axis, bar end, centre exclusion boundary, disc, and background sky.
- Paired evaluation of SEP and MTObjects using identical cases, folds, truth definitions, and metrics.
- Native-resolution products plus explicitly labelled derived views; the native FITS data remain authoritative.

Out of scope for the quantitative library

- APOD and amateur astrophotography images, because stretching, compositing, denoising, star reduction, PSF, calibration, and reuse rights are heterogeneous.
- Images cleaned by the algorithm under test and then treated as ground truth.
- Cross-survey images mixed into a single optimisation set without survey-specific PSF/noise modelling.
- A claim that the 40 clean galaxies are a statistically complete representation of all S4G morphologies; the pilot will measure coverage and identify gaps.

Core dataset contract

Product	Definition	Scientific use
clean_base.fits	Reviewed native S4G science image; never overwritten	Control image and damage measurement
foreground.fits	Added foreground flux only	Exact signal provenance and flux accounting
contaminated.fits	clean_base + foreground under a recorded seed	Input to SEP/MTObjects
truth_mask.fits	Binary foreground footprint under a declared truth threshold	Pixel recall, precision, leakage
truth_labels.fits	Integer source IDs and optional component classes	Per-source and failure-mode analysis
injections.csv	Position, flux, morphology, PSF, zone, seed	Reconstruction and stratified reporting
case.json	Paths, hashes, versions, units, WCS, configuration	Audit and reproducibility

3. Work plan

Phase 0 — Freeze the experiment specification (Week 1)

1. **Define "clean".** Use operational criteria: no visually significant foreground source within the measurement region, no unresolved saturation/ghost artifact, valid science data and geometry, and no prior algorithmic replacement used as truth.
2. **Define truth.** Choose explicit thresholds for the core, wings, and spike components. Keep both a strict truth mask and an expanded photometric-impact mask if they answer different questions.
3. **Define the primary endpoint.** Adopt bar-profile fidelity as the principal scientific endpoint, supported by mask recall/precision and clean-control damage metrics.
4. **Version the specification.** Store the agreed schema, eligibility rules, seeds, fold policy, and metric equations in a machine-readable configuration committed with the code.

Phase 1 — Audit and stratify clean S4G bases (Weeks 1–2)

Re-review the existing 40-galaxy list rather than assuming its filename suffix is sufficient validation. The audit should be blind to the preferred algorithm settings.

- Generate consistent review panels at native and stretched display scales, with bar geometry and science aperture overlaid.
- Record reviewer, date, status, contaminant notes, galaxy centre, pixel scale, PSF information, dimensions, and file checksum.
- Assign morphology/inclination/bar-size strata and quantify whether the 40 bases adequately cover the intended S4G science sample.
- Resolve disagreements explicitly; exclude questionable cases from v1.0 rather than silently accepting them.

Gate 1 Proceed only when every base has an immutable checksum, an eligibility decision, complete geometry, and a traceable visual-review record.

Phase 2 — Build the empirical foreground model (Weeks 2–4)

Construct a contaminant library appropriate to IRAC 3.6 µm data. Where possible, derive isolated-star stamps and PSF wings from S4G frames outside the target measurement region. Supplement these with parameterised models only where empirical coverage is insufficient.

- Separate ordinary point-source, bright-wing, saturated/near-saturated, and spike-bearing classes.
- Retain signed background-subtracted stamps, masks, normalization metadata, PSF orientation, and source provenance.
- Estimate realistic flux and spatial distributions from contaminated S4G fields rather than choosing uniform toy ranges by convenience.
- Include a small adversarial stratum crossing the bar major axis and bar ends; label it as deliberate oversampling.

Phase 3 — Implement the paired-case generator (Weeks 4–6)

- Extend the existing paired-toy manifest workflow instead of creating an unrelated pipeline.
- Generate all products from a single case seed; ensure reruns reproduce identical pixels and catalogue rows.
- Reject placements outside valid coverage and record every rejection reason.
- Preserve flux conservation during sub-pixel shifting and PSF rotation; test edge and NaN handling.
- Write atomic outputs to a versioned directory and verify hashes before marking a case complete.

Gate 2 A 20-case smoke set must reproduce bit-for-bit, pass FITS/WCS/schema checks, and show exact agreement between contaminated − clean and the stored foreground layer within numerical tolerance.

Phase 4 — Create folds and the 1,000-case pilot (Weeks 6–7)

Allocate galaxies—not individual realizations—to folds. All 25 realizations of a base galaxy must remain together. Stratify folds by morphology, inclination, bar size, and clean-image background characteristics as far as the sample permits.

Partition	Illustrative allocation	Permitted use
Training	24 galaxies / 600 cases	Parameter optimisation and diagnostic iteration
Validation	8 galaxies / 200 cases	Model/parameter selection and stopping decisions
Locked test	8 galaxies / 200 cases	One-time final comparison and reporting

The exact 24/8/8 split may be adjusted after the coverage audit, but the locked-test principle and galaxy-level grouping should not change after optimisation begins.

Phase 5 — Unified optimisation and evaluation (Weeks 7–10)

Primary metrics

- Bar-profile error: robust relative error and integrated absolute deviation within the defined bar measurement interval.
- Clean-control damage: fraction of unpolluted galaxy pixels masked or altered, plus flux/profile change on clean bases.
- Foreground recovery: truth-mask recall and precision, reported both by pixel area and by injected source.
- Residual contamination: unrecovered injected flux within the measurement aperture and bar-profile extraction strip.
- Failure severity: central/bar-crossing catastrophic failures reported separately from easy off-galaxy sources.

Selection rule

Use a constrained optimisation rather than a single unconstrained score: first require minimum recovery on scientifically important injections, then minimise clean-control damage and bar-profile error. Report medians, dispersion, worst-case tails, and bootstrap confidence intervals by held-out galaxy.

Gate 3 Freeze candidate SEP and MTObjects configurations before opening the locked test fold. Any later change creates a new experiment version and a new test set.

Phase 6 — Release, document, and decide scale-up (Weeks 10–12)

- Run the frozen configurations once on the locked test set and generate paired comparison panels.
- Publish the dataset manifest, generation configuration, folds, hashes, summary tables, failure gallery, and environment/dependency record.
- Write a methods note distinguishing simulated truth, visually clean controls, and qualitative real-field tests.
- Decide whether to expand the number of clean bases, increase realizations, add new foreground classes, or introduce an external-survey robustness set.

4. Proposed repository architecture

Keep large FITS products outside Git if necessary, but keep manifests, schemas, configurations, small QA fixtures, and code under version control. Paths below are conceptual and should be resolved through the existing machine-path abstraction.

Location	Contents
S4G_vNext/spec/	Dataset schema, clean criteria, truth definitions, metric specification
S4G_vNext/manifests/	Base-galaxy catalogue, foreground catalogue, fold manifest, release manifest
S4G_vNext/config/	Injection distributions, seeds, truth thresholds, evaluation settings
S4G_vNext/library/clean/	Immutable clean FITS bases and metadata sidecars
S4G_vNext/library/foreground/	Empirical/parameterised contaminant components
S4G_vNext/cases/<release>/	Generated paired cases and exact truth products
S4G_vNext/qa/	Review panels, schema reports, smoke fixtures, failure galleries
S4G_vNext/results/	Frozen run outputs, summaries, configuration snapshots

5. Roles, controls, and acceptance criteria

Workstream	Accountability	Acceptance evidence
Clean-base curation	Scientific reviewer	Signed review status, notes, checksum, geometry completeness

Workstream	Accountability	Acceptance evidence
Foreground modelling	Pipeline developer + reviewer	Empirical provenance, flux/PSF distributions, visual stamp QA
Case generation	Pipeline developer	Deterministic smoke tests, conservation tests, schema validation
Fold design	Analysis owner	Galaxy grouping, stratification report, frozen fold hash
Optimisation	Method owner	Config snapshots, repeatable runs, no test-fold access
Final evaluation	Analysis owner	Locked run, uncertainty estimates, failure analysis
Release	Project owner	Version tag, manifest, documentation, checksums, archive location

Definition of done for S4G vNext pilot

- All 40 candidate bases have auditable status; only accepted bases appear in the released fold manifest.
- At least 1,000 valid cases are generated, or the final count and exclusions are transparently documented.
- Every case reconstructs from its manifest and passes flux-layer consistency checks.
- SEP and MTObjects are evaluated on precisely the same cases and truth masks.
- No galaxy identity crosses train/validation/test boundaries.
- The locked test is opened only after configuration freeze and is not recycled for tuning.
- A third party with the S4G inputs can reproduce manifests, injections, and summary metrics from documented commands.

6. Principal risks and mitigations

Risk	Consequence	Mitigation
"Clean" bases contain faint contaminants	Truth is biased and clean-damage estimates are unreliable	Two-pass visual audit; Gaia/SEP-assisted flags; exclude uncertain cases
Toy contaminants are unrealistic	Optimised settings fail on real S4G images	Use empirical IRAC stamps/distributions and a separate real-field stress test
Fold leakage	Performance is overstated	Group all realizations by base galaxy before stratification
Single aggregate score hides damage	High recovery wins by masking galaxy structure	Constrained selection plus separate damage/profile metrics
Truth-mask threshold is arbitrary	Results depend on annotation convention	Publish strict and impact masks; run threshold sensitivity analysis
Large storage footprint	Slow iteration and difficult sharing	Immutable bases; generated release cache; compressed FITS; checksummed manifests
Parameter overfitting to S4G	Limited external validity	Reserve a later external-survey robustness set; never merge it into S4G tuning

7. Twelve-week pilot schedule

Weeks	Milestone	Key output / decision
1	Specification freeze	v0.1 schema, clean definition, truth definition, metric contract
1–2	Clean-base audit	Accepted base catalogue and coverage-gap report
2–4	Foreground library	Versioned empirical/parameterised source components
4–6	Generator implementation	Deterministic paired-case generator and 20-case smoke set
6–7	Pilot generation	1,000 cases and frozen galaxy-level folds
7–10	Optimisation/validation	Frozen SEP and MTObjects candidate configurations
10–11	Locked test	Final comparative metrics and failure analysis
12	Release decision	v1.0 pilot package and scale-up recommendation

8. Immediate next actions

1. **Approve the dataset contract.** Confirm that the authoritative unit is clean base + foreground layer + contaminated image + exact truth + manifest.
2. **Audit the current 40.** Create a review panel and machine-readable base catalogue from CleanGalaxies.txt and the S4G geometry manifest.
3. **Inventory existing toy code.** Map the current paired-manifest generator, injection routines, evaluators, and machine paths against the vNext schema; reuse before rewriting.
4. **Build a 20-case vertical slice.** Use two galaxies, five source classes/locations, and two seeds to validate the complete path from generation through SEP/MTObjects metrics.
5. **Review evidence before scaling.** Only launch the 1,000-case pilot after deterministic reconstruction, flux consistency, and visual truth overlays pass.

References and methodological basis

Haigh, C. et al. (2021). Optimising and comparing source-extraction tools using objective segmentation quality criteria. Astronomy & Astrophysics, 645, A107. https://doi.org/10.1051/0004-6361/201936561

Project basis. Existing repository workflows include the 40-galaxy CleanGalaxies.txt convention, paired toy-manifest generation, galaxy-held-out cross-validation inputs, SEP processing, MTObjects spike-gate processing, and report generation.